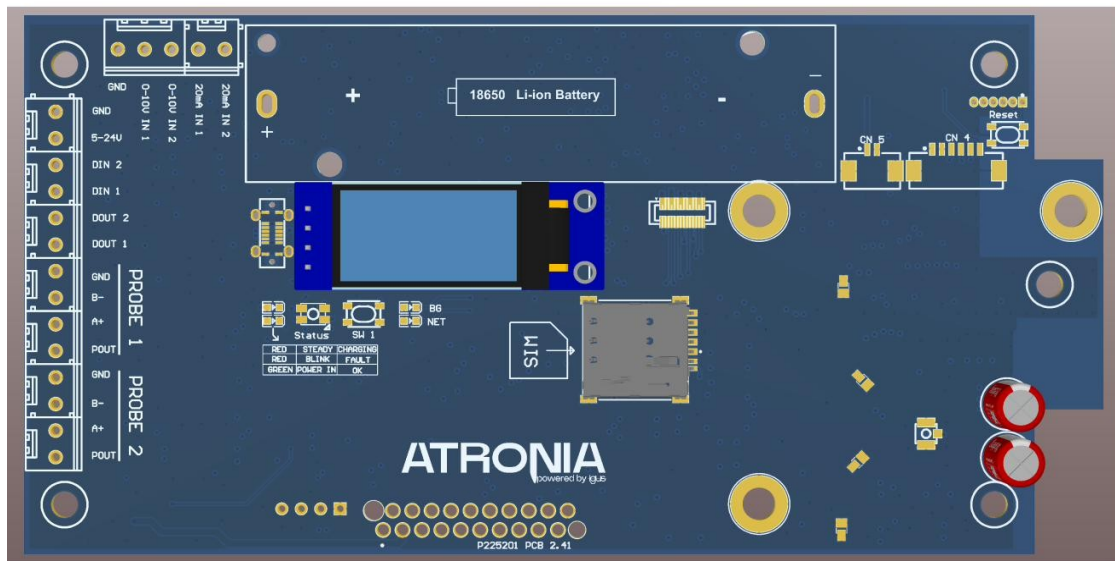


SensBlue Monarch

Cellular IoT that fits any field. Any climate. Any grid.



Cellular first

LTE-M / NB-IoT / EGPRS fallback and integrated GNSS — no gateway needed for remote sites.

Solar-ready

Multi-source input auto-selects USB, battery, external DC or solar panel. Built-in Li-ion charging.

Field-grade I/O

Isolated digital I/O, 0-10 V and 4-20 mA analog inputs, RS485 with 15 kV ESD protection.

Arduino ready

Program from the Arduino IDE. 12 open-source examples cover every on-board peripheral.

Enclosure aware

Board mounts directly into Gainta G258 IP65/IP67/IP68 enclosures. Solar lid available as a kit.

Open by design

Schematics, library and examples on GitHub under the MIT licence. Yours to adapt.

Where it fits

The SensBlue Monarch was designed for deployments where cellular connectivity, low power, and industrial-grade I/O are simultaneously required. A single device handles the whole edge from field signals to the cloud.

Smart agriculture Soil, irrigation and crop-microclimate monitoring on solar power. LoRa-adjacent range through NB-IoT, without the gateway.	Smart cities Waste, parking, noise and air-quality gateways with reliable LTE-M coverage inside metal cabinets and underground.
Perimeter security Solar-park fence-line and industrial-site intrusion detection with isolated inputs for fence sensors and outputs for beacons.	Environmental telemetry River, weather and pollution monitors that survive winter on a Li-ion 18650 and a small solar lid.
Asset & fleet tracking GNSS out-of-the-box, LTE-M for reliable position reporting even at power thresholds where LTE would fail.	Industrial signal conditioning 0-10 V or 4-20 mA sensor front-ends, RS485 daisy-chains, connected directly to your cloud via cellular.

At a glance

MCU	Espressif ESP32-WROOM-32E-N8, dual-core LX6 @ 240 MHz, 8 MB flash, Wi-Fi 802.11 b/g/n, BLE 4.2
Cellular	Quectel BG95-M3: LTE-M Cat.M1, NB-IoT Cat.NB1/NB2, EGPRS fallback
Positioning	Multi-constellation GNSS: GPS, GLONASS, BeiDou, Galileo (shared LTE antenna)
Sensing	Bosch BME280 (temperature, humidity, pressure) on I ² C; 256-byte AT24C02 EEPROM
Display	I ² C SSD1306 OLED header, 128×32 default (128×64 supported)
Digital I/O	2 × isolated inputs (EL357N, up to 30 V), 2 × high-side outputs (up to 500 mA)
Analog inputs	2 × software-selectable 0-10 V or 4-20 mA; shared mode across channels
Serial	RS485 half-duplex (ISL83483, 15 kV ESD), USB-UART via CP2102N
Power in	USB (5 V), Li-ion battery (2.5-4.4 V), external DC (up to 5.5 V), solar panel (up to 6.5 V)
Power out	Regulated 3.3 V for peripherals; 12 V boost for 4-20 mA loop and external use
Charging	TI BQ25176 Li-ion charger with power-path management; runs even with depleted battery
Programming	Arduino IDE via Espressif ESP32 core; open-source SensBlueMonarch library on GitHub
Dimensions	160 × 80 × 55 mm inside Gainta G258 enclosure; four M3 mounting holes
Temperature	-25 °C to +75 °C operating (extended range with low-temperature 18650)

Ecosystem

Gainta G258 enclosure IP65 / IP67 / IP68 variants. Opaque or transparent lid.	Solar panel lid kit 3.3 W / 6 V panel factory-mounted on a transparent-lid enclosure.	Antennas Adhesive PCB antenna, high-gain N-type, U.FL-to-N pigtail.	Low-temperature 18650 Li-ion cell rated -40 °C discharge for winter outdoor use.
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Get started

Everything you need to evaluate, prototype and deploy is one click away. The board and its ecosystem are supported by open-source drivers, reference examples and full schematics.

For developers

- ◆ Board support and library: github.com/sensblue-monarch/SensBlueMonarch
- ◆ 12 reference examples covering every on-board peripheral
- ◆ 11 schematic sheets (JPG) under [docs/schematics/](#)
- ◆ Detailed pinout: [docs/pinout.md](#)
- ◆ Espressif ESP32 core, Adafruit BME280 / SSD1306 supported
- ◆ MIT licence throughout

For buyers

- ◆ Board only: contact ATRONIA for pricing and MOQ
- ◆ Enclosure kit: Gainta G258MF / G258CMF / G258MF-IP67
- ◆ Solar lid kit: G258CMF with pre-mounted 3.3 W panel
- ◆ Antennas: adhesive PCB (U.FL), N-type high gain, pigtails
- ◆ Low-temperature 18650 cells for extended-cold deployments
- ◆ Full technical datasheet available on request

Talk to us

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